PATHI MANIKANTA

AI Evaluation Specialist | ML Engineer | RLHF/RLAIF

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PROFESSIONAL SUMMARY

AI Evaluation Specialist and Machine Learning Engineer with hands-on experience in building end-to-end ML pipelines, evaluating frontier AI systems, and contributing to RLHF/RLAIF workflows. Skilled in Python, scikit-learn, TensorFlow/Keras, NLP, computer vision, and agentic tool-use evaluation. Earned promotion from attempter to reviewer at Scale AI (Outlier) for maintaining 95%+ grading consistency across 100+ reviews. Built classification models achieving 92.5% accuracy with proper data leakage detection and GroupKFold cross-validation.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, SQL, HTML, CSS, Embedded C

ML & AI Frameworks: scikit-learn, TensorFlow, Keras, pandas, NumPy, NLTK, OpenCV, matplotlib, seaborn

AI Evaluation: LLM evaluation, RLHF, RLAIF, rubric design, agentic trajectory review, tool-use evaluation, API evaluation, red-teaming, prompt engineering, pytest verifiers, fact-checking, data annotation

ML Techniques: Classification, regression, model training, model evaluation, feature engineering, data preprocessing, signal processing, cross-validation, GroupKFold, hyperparameter tuning, data leakage detection, NLP, TF-IDF, text classification, computer vision, face detection, face recognition

Tools & Platforms: Git, GitHub, MATLAB, Excel, Power BI, Arduino, AWS (EC2, S3, IAM, VPC), Google Workspace, VS Code, Jupyter, pytest

Web & Frameworks: React, Node.js, REST APIs, JSON, HTML/CSS

Spoken Languages: English, Telugu, Hindi, Tamil

PROFESSIONAL EXPERIENCE

AI Evaluation Specialist, Freelance | Scale AI (Outlier) - OpenClaw Atlas, Blue Shell & Lobster Safety | **2026 - Present**

- Promoted from attempter to reviewer within 3 months after designing single-turn agentic evaluation tasks with consistent grading accuracy and full guideline compliance.
- Authored 50+ binary pass/fail rubrics (15-25 criteria each) and developed pytest verifiers that programmatically validate task outputs using systematic test design methodology.
- Audited 100+ agentic tool-use trajectories, identifying failure patterns in call sequencing, argument accuracy, and instruction-following to improve frontier model performance.
- Architected multimodal evaluation tasks across 10+ categories, enforcing cross-modal reasoning and multi-step tool coordination for frontier AI systems.
- Contributed to the RLHF/RLAIF feedback loop training frontier AI models, pinpointing hallucination patterns, instruction-following failures, and incorrect tool call execution.
- Authored reusable end-to-end task workflows (story to prompt to silver nudge to rubric to pytest verifiers to QC) across OpenClaw Atlas, Blue Shell, and Lobster Safety, and published a public workflow walkthrough and an automated QC-checker tool.

PROJECTS

Diabetes Detection Using Foot Pressure | Final-Year B.E. Project, Anna University | **2025**

- Engineered an end-to-end ML pipeline: collected data from 6 FSR sensors via Arduino across 30+ walking sessions at 20Hz, processed signals using Butterworth low-pass filter in MATLAB, and trained classifiers in Python.
- Identified and resolved session-level data leakage inflating accuracy from 99% to a realistic 92.5% by implementing GroupKFold cross-validation, demonstrating production-level ML debugging skills.
- Extracted 80+ features (statistical, pressure distribution, temporal) and benchmarked 4 models: Random Forest (91.2%), SVM (89.7%), Logistic Regression (87.4%), Keras Neural Network (92.5%).

Spam Email Classifier | NLP Project | **2024**

- Developed a complete NLP classification pipeline: text preprocessing with NLTK (stemming, stopword removal), TF-IDF vectorisation (5000 features, unigrams + bigrams), achieving 97.8% accuracy with Linear SVM on 1000+ emails.
- Benchmarked Naive Bayes, Logistic Regression, and SVM using precision, recall, F1-score, and confusion matrix analysis for systematic model selection.

Real-Time Face Recognition | Computer Vision Project | **2024**

- Implemented a real-time face detection and recognition pipeline using OpenCV (Haar Cascades + LBPH recognizer), processing live video at 15+ FPS with 70%+ confidence threshold on a custom dataset of 50+ images per person.

IoT Environmental Monitoring System | Vaayusastra Internship | **2023**

- Deployed a real-time sensor data pipeline using Arduino (DHT11, BMP180) streaming temperature, humidity, and pressure readings every 2 seconds to a remote dashboard, handling noisy real-world sensor data.

EDUCATION

Bachelor of Engineering, Computer Science & Engineering (Data Science)

Jansons Institute of Technology (Anna University), Coimbatore | 2021-2025 | CGPA 8.0/10

Higher Secondary (Class XII), MPC

Sri Sai Junior College, Kanigiri, Andhra Pradesh | 2021

Secondary (Class X)

Nagarjuna Model School, Kadapa, Andhra Pradesh | 2019 | CGPA 9.8/10

CERTIFICATIONS

AWS Academy Cloud Foundations, Amazon Web Services, 2022

Microsoft 365 Productivity Suite (Advanced), Microsoft, 2022

Ethical Hacking & Penetration Testing, Scholiverse Educare Pvt. Ltd.